CONFLUENTIA × 2UNE.AI RL HACKATHON • TRACK: SOFTWARE & DEVOPS

OpsArena: Production Cloud-Native SRE Environment

A Hardened, Deterministic Reinforcement Learning Environment for Autonomous Incident Response

8 / 8 TASKS PASSED Pre-flight baseline checks verified in 2.53s	**0.9955 LIVE SCORE** Autonomous gpt-6-luna incident rollback in 13.7s	**ZERO-TRUST SANDBOX** Unprivileged UID 1000, internal-only net, no host socket	**SINGLE COMMAND** docker compose up --build ... exit code forward

Author & Team: team_087 • **Evaluated Model Deployment:** gpt-6-luna (penpencil.guru) • **Task ID:** task_02_bad_deployment

The Challenge: SRE Benchmarking for Autonomous RL Agents

Bridging the Chasm Between Synthetic Toy Problems and Fragile Sandboxes

Current Landscape Flaws

- **Synthetic Toy Problems:** Most benchmarks test math or simple games with zero real-world infrastructure relevance.
- **Fragile Docker Sandboxes:** Real-world bash sandboxes frequently mount host sockets, leak environment credentials, or crash.
- **Prompt Injection & Cheating:** Unconstrained agents rewrite test files, mock verifier functions, or leak hidden answers.
- **Binary All-or-Nothing Scoring:** Agents receive 0% for making 90% diagnostic progress, discouraging nuanced exploration.

OpsArena's Engineering Solution

- **Production CloudMesh Simulator:** 8 interdependent microservices modeling cascading failures, pool exhaustion, and memory leaks.
- **Strict Dual-Container Isolation:** Zero internet egress for sandbox, unprivileged UID 1000, no Docker socket, 512MB RAM ceiling.
- **Hardened Action Allowlist:** No arbitrary bash/python exec; only 8 typed, schema-validated operational tools.
- **Bounded Continuous Partial Credit:** Mathematically gated reward providing smooth optimization without breaking no-op constraints.

Dual-Container Architecture & Isolation Topology

Separation of Powers: Verifier Inaccessibility and Strict Process Containment

MODEL API LAYER
penpencil.guru (gpt-6-luna)

→ HTTPS Egress →
model prompts & tool responses

HARNESS CONTAINER (Evaluator)
Dual-homed: 'internet' + 'isolated'
- Pre-flight gate orchestration
- Context trimming (48k tokens)
- Multi-factor programmatic verifier

→ HTTP /tool, /reset, /state →
Internal docker bridge only

SANDBOX CONTAINER (CloudMesh)
Isolated network only (internal: true)
- Unprivileged user (UID 1000)
- Zero internet egress (no gateway)
- 512MB RAM, 256 PIDs, cap_drop: ALL
- Output strictly clipped at 4k chars

Key Security Defense	Implementation in OpsArena	Adversarial Attack Prevented
Verifier Inaccessibility	Verifier code resides strictly inside the harness; sandbox exposes only /state.	Agent tampering with reward calculation or ground truth labels.
Zero Host Socket Access	/var/run/docker.sock is never mounted; cap_drop ALL; no-new-privileges.	Host escape, container breakout, and spawning rogue sibling containers.
Automated Secret Scrubbing	In-flight logs and telemetry regex-scrub database passwords, tokens, and keys.	Agent exfiltrating or leaking credentials in tool query results.

The CloudMesh Simulator: 8 Enterprise Microservices

Realistic Dependency Topologies, Dynamic Metrics, and Degradation Propagation

Service	Tier	Default Pods	Dependencies	Operational Dynamics & Incident Roles
api-gateway	Edge	3 pods	order, recommendation	Ingress rate limiting, circuit breaking, SSL termination, and client routing.
order-service	App	3 pods	payment, inventory, cache	Core transaction pipeline; subject to bad releases, canary leaks, and traffic surges.
payment-service	App	2 pods	db-primary, 3rd-party	Outbound payment partner integration; circuit breaker isolation under upstream latency.
inventory-service	App	2 pods	db-primary, db-replica	Database connection pooling; subject to connection pool starvation under checkout bursts.
recommendation	App	2 pods	cache-cluster	Soft dependency; fails gracefully to fallback recommendations if degraded.
cache-cluster	Cache	2 pods	None	Redis memory eviction storm; capacity scaling prevents thundering herd to DB.
db-primary	Data	1 pod	None	PostgreSQL write primary; table lock contention and migration lock timeouts.
db-replica	Data	2 pods	None	Read replica pool; overprovisioning causes financial drain without adding throughput.

Unified Physics: compute_service_metrics serves as the single source of truth for both model queries and verifier evaluation, guaranteeing 100% agreement between reported observations and final scoring.

The 8-Task Incident Suite with Seeded Diversity

Deterministic, Reproducible Scenarios with Procedural Parameter Perturbation

ID	Incident Title	Diff	Seeded Perturbations	Target SRE Mitigation	Oracle	No-Op
task_01	Redis Eviction Storm	Easy	Mem capacity (1-4GB), surge QPS (2.8k-3.6k)	Scale cache-cluster ≥ 2 replicas	**0.996**	**0.000**
task_02	Bad Deployment Rollback	Easy	Bad release tags, error rates (16-21%)	Roll back order-service to v2.4.0	**0.996**	**0.000**
task_03	DB Pool Starvation	Med	Initial pool (3-8), checkout RPS (1.6k-2.2k)	Update db_pool_size ≥ 25	**0.996**	**0.000**
task_04	Slow Third-Party Gateway	Med	Partner hang (6s-9s), thread exhaustion	Enable circuit_breaker_enabled	**0.996**	**0.000**
task_05	Cascading Failure	Hard	Flash sale surge (4.4k-5.2k RPS), OOM loops	Scale order-service ≥ 6 replicas	**0.994**	**0.000**
task_06	Canary Memory Leak	Med	Canary split (5-15%), leak rate	Reroute canary traffic weight to 0%	**0.996**	**0.000**
task_07	Overprovisioned Drain	Hard	36 pods burning $540/hr under 12% CPU	Right-size cluster to ≤ 18 pods	**0.996**	**0.000**
task_08	Migration Lock Starvation	Hard	Blocked table, migration PID (4200-8900)	Set lock_timeout_ms to 1000ms	**0.996**	**0.000**

Seed Invariance: The seed perturbs operational conditions to prevent rote memorization while ensuring every variation is fully solvable by the reference oracle (≥ 0.95) and deterministic across resets.

Operational Action Space & Key-Specific Schema Defense

8 Typed Tools with In-Depth Parameter Type and Value Range Enforcement

Tool Name	Cost	Allowed Parameters	Adversarial Defense & Key-Specific Enforcement
`get_topology`	$1.00	None	Returns cluster state, replicas, versions, and budget spent without exposing hidden tokens.
`query_metrics`	$2.00	service, metric (latency, errors, cpu, memory, qps, rps)	Full alias support (QPS/RPS); returns 5-step timeseries with status indicator.
`query_logs`	$3.00	service, filter (optional regex)	Regex secret scrubbing replaces database passwords, JWT tokens, and API keys with [REDACTED].
`scale_service`	$2/pod+$3	service, replicas (int: 1 to 20)	Rejects non-ints, booleans, and out-of-range scaling requests before state modification.
`rollback_release`	$20.00	service, target_version (string)	Strictly validates target version against release registry; rejects arbitrary version injection.
`update_config`	$5.00	service, key, value	**Key-specific typed schema:** Rejects 'banana' on db_pool_size; enforces int ranges and booleans.
`reroute_traffic`	$10.00	service, weight (int: 0 to 100)	Enforces integer bounds (0-100); prevents negative weights or string injections.
`finish`	$0.00	None	Signals episode completion; triggers formal programmatic scoring.

Budget Enforcement: Cumulative spend is tracked dynamically. If an agent attempts an action that would exceed the $150.00 operational budget, the simulation rejects the request, protecting the cluster.

Programmatic Reward Formulation & Partial Credit

Mathematical Zero-Tolerance No-Op Gate with Continuous Diagnostic Progress Scoring

Continuous Gated Formulation: $\text{Reward} = P \times [\, 0.45 \times S_{SLO} + 0.25 \times R_{reg} + 0.15 \times B_{bud} + 0.15 \times E_{eff} \,]$

Factor	Weight	Scoring Dynamics	Mathematical Contract & Safeguard
P (Progress Gate)	Multiplier	P = 0.00 for no-op/idle; P ∈ (0, 0.5] for partial progress; P = 1.00 for full resolution.	Mathematically enforces that no-op agent scores **0.0000** ≤ **0.05** while rewarding partial fixes.
S_SLO (SLO Recovery)	45%	Scores post-incident error rate (≤ 0.1%) and p99 latency (≤ 100ms).	Penalizes sluggish response or incomplete error mitigation.
R_reg (Platform Stability)	25%	Evaluates non-target microservices against their pre-incident baseline health.	Deducts heavily if an agent damages, downscales, or misconfigures unrelated services.
B_bud (Budget Preservation)	15%	Full marks for spend ≤ $50.00; decays linearly to 0 at the $150.00 ceiling.	Penalizes wasteful, brute-force scaling or unnecessary rollbacks.
E_eff (Action Efficiency)	15%	Full marks for resolution within ≤ 8 tool turns; decays linearly up to 40 turns.	Rewards surgical diagnosis over aimless polling loops.

Empirical Results: Baseline Gating & Live gpt-6-luna

Verified Evidence: 8/8 Tasks Pre-Flight Pass + Autonomous Incident Resolution

1. Pre-Flight Baseline Checks (gate.json)

- **Pre-flight Check Duration: 2.53 seconds** (budget: 120s)
- **Oracle Solutions:** All 8 tasks score **0.994 – 1.000** (≥ 0.95 PASS)
- **No-Op Agents:** All 8 tasks score exactly **0.0000** (≤ 0.05 PASS)
- **Multiple Seeds Tested:** Seed 0, 42, 99 all pass with 0 errors
- **Gate Enforcement:** Exit code 2 if either check fails

2. Live Model Run (gpt-6-luna on task_02)

- **Turn 1 (Triage):** Polled topology, metrics, logs; identified NPE in v2.4.1
- **Turn 2 (Remediate):** Rolled back order-service to stable v2.4.0
- **Turn 3 (Verify):** Confirmed error rate dropped from 18.4% → 0.04%
- **Turn 4 (Close):** Called finish; final reward **0.9955 / 1.0000**
- **Wall-Clock Elapsed: 13.7 seconds** (limit: 600s)

Sample Trajectory Evidence	Task ID	Turns	Reward	Evaluation Status & Key Finding
trajectory_success_task_02.md	task_02	4 turns	**0.9955**	SUCCESS: Clean triage → rollback → verification → sign-off in 13.7s.
trajectory_failed_task_05.md	task_05	1 turn (cap)	**0.0000**	FAILED (Negative Control): Step cap reached during read phase before scaling.

Adversarial Threat Model & Tested Security Controls

Empirically Verified Defenses Against Agent Exploitation and Cheating

Adversarial Vector	Exploit Mechanism	OpsArena Defense Mechanism	Verification Test
Malformed Config Crash	Passing 'banana' or -500 to update_config crashing verifier.	Key-specific schema validation; rejects invalid types before state storage.	`test_malformed_config_inputs_do_not_crash_scorer (PASS)`
Context-Window Stuffing	Agent triggers huge log dumps to overflow context or force crashes.	Output clipping at 4,000 characters; sliding context trimming budget at 48k tokens.	`test_output_clipping (PASS)`
Unrelated Service Damage	Agent solves incident by wrecking other services or draining pods.	Dynamic regression scoring compares non-target services to baseline health.	`test_dynamic_regression_penalty (PASS)`
Prompt Injection in Logs	Malicious payload in log payload instructing agent to self-certify.	Verifier is hosted in harness; self-certification commands do not affect score.	`Architectural separation (dual-container)`

Honest Threat Model: While process breakout and reward tampering are prevented by design, future iterations will expand from discrete-event simulation to multi-node Kubernetes clusters with real network packet injection.

Turnkey Submission Checklist & Reproduction

100% Contract Compliance with Single-Command Evaluation Workflow

Submission Deliverable	Required Path in team_087.zip	Compliance Status & Verification
Single-Command Compose	docker-compose.yml	VERIFIED: Builds sandbox on isolated net and harness on internet.
Baseline Gate Report	baseline-results/gate.json	VERIFIED: All 8 tasks pass (Oracle: 0.996, No-Op: 0.000) in 2.53s.
Live Model Evaluation Logs	test-run-logs/	VERIFIED: 3 genuine gpt-6-luna runs (run_01, run_02, run_03).
Sample Trajectories	sample-trajectories/	VERIFIED: 1 success trajectory (task_02) + 1 failed trajectory (task_05).
Technical Design Document	design-doc.pdf	VERIFIED: 2-page dense, publication-quality technical specification.
Presentation Slide Deck	deck.pdf	VERIFIED: 10-slide keynote presentation deck.
API Key Protocol	.env (excluded)	VERIFIED: .env in .gitignore, excluded from team_087.zip.

One-Command Evaluation: `docker compose up --build --abort-on-container-exit --exit-code-from harness`

OpsArena is 100% complete, verified, and ready for judging. Thank you!